Identifying Dangerous Migrations

As an experienced Rails developer and PostgreSQL user, you've likely had Migrations that didn't run correctly and possibly blocked your release process. Blocking releases slows down your development team velocity. Besides slowing the team down, Migrations could even cause application errors.

One solution would be to take your database down when you need to make schema changes. With a disconnected database there would be no concurrent modifications to worry about from application queries, so changes would be perfectly safe!

Unfortunately that strategy is impractical. Modern development teams don't take their databases offline to make changes. Modern teams are shipping code changes and evolving SQL database schemas every day, and fortunately PostgreSQL can keep pace with the needs of your team, allowing you to change the structure as often as needed.

How do you detect those scenarios?

When a developer creates an Active Record Migration for a DDL change, there is no built-in concept of safety or danger with Migrations themselves. All migrations are treated equally. This leaves the possibility open of deploying unsafe migrations that can harm team velocity and cause application errors.

Fortunately third party gems like Strong Migrations[2] fill this need. Strong Migrations has been added to Rideshare so you can experiment with it there. The main purpose of Strong Migrations is to identify potentially unsafe migrations during normal development, using well known patterns from DDL changes that can cause trouble. Strong Migrations even presents safer alternatives.

Read on to learn more.

Learning From Unsafe Migrations

To get some hands-on experience with how queries can get blocked, you'll create a long running modification and run a query during that time that gets blocked. You'll use the Rideshare database and the `temp` schema from earlier.

Launch psql from your terminal, connecting to the Rideshare database. To reset the `temp` schema, use the `CASCADE` option which drops any tables in that Schema. Create it again.

```
DROP SCHEMA IF EXISTS temp CASCADE;
```

2. https://github.com/ankane/strong_migrations